Disclosure Without Inference

LLM Agents Anchor on Disclosed Trade Data When Bargaining

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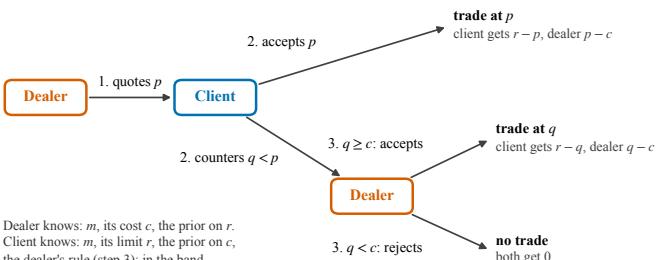
Motivation and research question

- AI agents are starting to negotiate trades for people, and one side may field a far more capable model. Which protocols still protect the weaker side? (AGI Institutions)
- One proposal: an information floor, giving the weaker side information it cannot infer itself. When US corporate bond trades were published (TRACE), investors' trading costs fell (Goldstein et al., 2007).

1. Does disclosing other dealers' trade data improve an LLM client's outcome when it bargains with a dealer?
2. Does the client use the disclosure as a Bayesian would, or merely anchor on its numbers?

Setting: a client buys $10mm of an off-the-run 10-year US Treasury note from a single dealer whose cost is private. Off-the-run trades are reported to FINRA but not published: the band models publishing them.

The game



Notation

m	evaluated price: an indicative price both sides see
c	dealer's cost; private to the dealer
r	client's limit, the most it will pay; private to the client
p	dealer's quote
q	client's counteroffer
g	price unit: 1/10 of daily price volatility (about \$3,400 on \$10mm)
band	disclosed range for the dealer's cost, from 10 other dealers' trades

Prices are measured in g, one tenth of the note's daily price volatility (about \$3,400 on \$10mm). Outcome: the client's share of the surplus, $(r - \text{price})/(r - c)$, or 0 if no trade.

The band, the Bayesian benchmark and the design

In some conditions the client also reads:

> *"Reported purchases by other dealers of similar off-the-run notes yesterday (10 trades) . . . imply a typical dealer cost of avg. . . . this dealer's own cost is within [low, high] about two times in three."*

- A noisy but accurate reading of the dealer's cost.
- **Bayesian client**: combines the band with its prior and bids optimally. Against the same fixed dealer, the band raises its share from 62% to 73%.

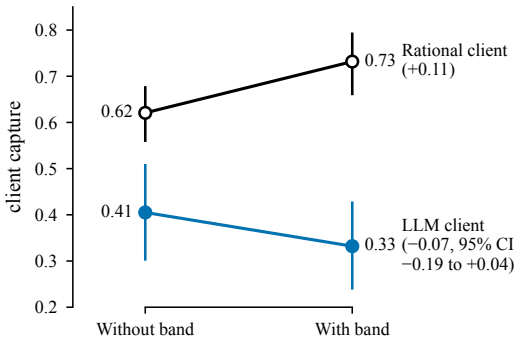
2 × 2 conditions

	No band	Band
No belief elicitation	✓	✓
Belief elicitation	✓	✓

- Elicitation: P(cost below each of nine prices), before and after the quote.
- Client: Ornith-1.5-35B, temperature 0.5, reasoning off; chosen by a pilot fixed in advance.
- 50 scenarios × 2 runs; main test pre-registered; 20 held-out scenarios.

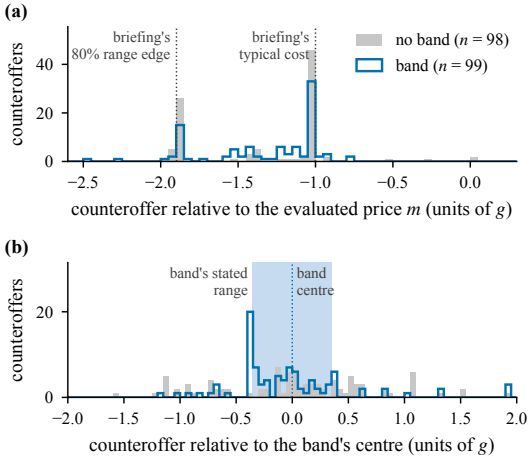
The most a client that uses the band perfectly can gain: 11% of the surplus

Result 1: the band did not help



- Without belief elicitation: 41% → 33% of the surplus.
- Difference −7.3 points, 95% CI [−19.0, 3.8]: main test not supported.
- Paired test: the LLM gained 18 points less than the Bayesian client (95% CI [5, 32], $p \approx 0.008$).

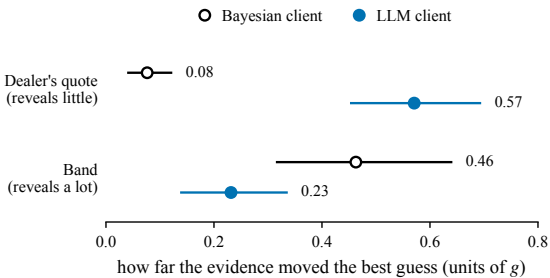
Result 2: the client bids the numbers it is given



- Without the band: 77 of 98 bids sit on two prices from the briefing.
- With the band, bids move towards its centre ($0.16g$, pre-registered).
- 26% sit on the band's lowest price; the cost is below it only ∼16% of the time, so 77% were rejected.
- Belief elicitation's own prices became anchors too.

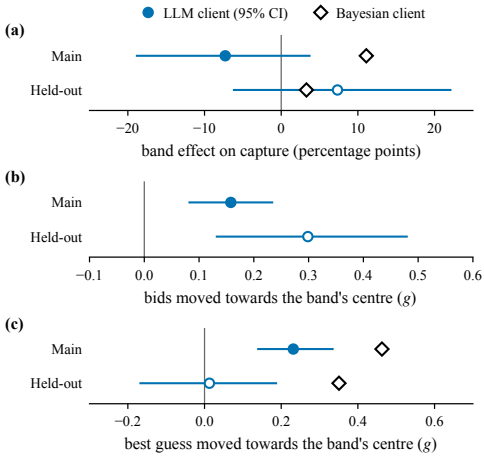
Anchoring, not inference

Result 3: it weighs evidence the wrong way round



- The **quote** reveals almost nothing: the dealer quotes the same price whatever its cost.
- Yet it moved the LLM's best guess $0.57g$, against $0.08g$ for a Bayesian.
- The **band** reveals a lot, yet moved it only $0.23g$, against $0.46g$.
- Its bids barely followed its own stated beliefs.

Confirmation on unseen scenarios



- 20 scenarios never used in development; analysis code unchanged.
- (a) Again no reliable effect on the client's share.
- (b) Again bids move towards the band's centre: anchoring replicates.
- (c) Beliefs again lag the Bayesian's, and on these scenarios do not move at all.

Limitations

- **Simplified market**: one dealer, one counteroffer, walking away worth nothing.
- **Fixed dealer**: it neither adapts to nor exploits the client, and its quote reveals little.
- **Synthetic values**: only the price scale is calibrated to market data.
- **One model**, reasoning off, one prompt.
- The briefing states the cost distribution as **concrete prices**, which themselves became anchors.

Next: reasoning on; other models and sizes; the briefing without price levels.

Takeaways

1. Disclosure cannot substitute for inference the agent does not perform. The LLM noticed the band but bid its numbers.
2. Presentation matters as much as disclosure. An anchoring agent bids whatever number it is shown.
3. Asking about beliefs is not neutral. The question's own prices became bids.
4. Information floors alone may not protect weaker agents: marketplaces may also need minimum capability standards, which games with a Bayesian benchmark could help test.

Thank you. Questions?

AGI Institutions Wiki (2026). Agent negotiation & bargaining. Goldstein, Hotchkiss & Sirri (2007). *Review of Financial Studies* 20(2). Myerson & Satterthwaite (1983). *Journal of Economic Theory* 29(2). Tversky & Kahneman (1974). *Science* 185(4157).